arXiv:2602.16740v1 [cs.LG] 17 Feb 2026

Quantifying LLM Attention-Head Stability: Implications for Circuit Universality

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Abstract

In mechanistic interpretability, recent work scrutinizes transformer "circuits"—sparse, mono or multi layer sub computations, that may reflect human understandable functions. Yet, these network circuits are rarely acid-tested for their stability across different instances of the same deep learning architecture. Without this, it remains unclear whether reported circuits emerge universally across labs or turn out to be idiosyncratic to a particular estimation instance, potentially limiting confidence in safety-critical settings. Here, we systematically study stability across-refits in increasingly complex transformer language models of various sizes. We quantify, layer by layer, how similarly attention heads learn representations across independently initialized training runs. Our rigorous experiments show that (1) middle-layer heads are the least stable yet the most representationally distinct; (2) deeper models exhibit stronger mid-depth divergence; (3) unstable heads in deeper layers become more functionally important than their peers from the same layer; (4) applying weight decay optimization substantially improves attention-head stability across random model initializations; and (5) the residual stream is comparatively stable. Our findings establish the cross-instance robustness of circuits as an essential yet underappreciated prerequisite for scalable oversight, drawing contours around possible white-box monitorability of AI systems.

1. Introduction

Modern deep neural networks (DNNs) exhibit a notable phenomenon: DNNs trained at different starting initializations frequently yield solutions with similar prediction performance (Choromanska et al., 2015; Garipov et al., 2018; Scimeca et al., 2022). Yet, this begs a neglected corollary question today: do current transformers learn radically different sets of intrinsic computational schemes that happen to perform similarly, or do transformers learn internal representations that are largely the same? (Chughtai et al., 2023)

Further, the application scenarios explored in current large language model explainability research (LLM XAI) tend to be limited to narrow tasks and lack generalizability. The extracted "LLM circuits" are rarely validated in unseen settings or tasks (Olah, 2022). Some authors argue (Hendrycks & Hiscott, 2025) that current AI systems are too complex and that a top-down interpretability approach that captures emergent properties is preferable to attempts to reverse-engineer models "neuron by neuron" or in small functional blocks of neurons called "circuits". In other words, by focusing on representations as the primary units of analysis—rather than neurons or small circuits—this approach seeks meaning in patterns of activity across many neurons at a time. Furthermore, existing mechanistic research into transformers often relies on analyzing single attention heads in isolation, effectively cherry-picking learned modules without establishing their representativeness.

These observations can be seen through the lens of the "universality hypothesis" from the mechanistic interpretability literature (Chughtai et al., 2023; Olah et al., 2020). This hypothesis asserts that models learn the same or similar representational features and circuits across different models when trained on similar tasks. However, the strength of the similarities between these learned model components has yet to be systematically quantified, leaving the critical distinction between strong and weak universality unresolved. (Chughtai et al., 2023). Hence, in this work we make the notion of universality experimentally accessible by investigating whether the model converges to the requisite representational space—alignment of the attention head's induced representation, that is, the subspace it attends to and writes into—and therefore to the same functional behavior.

Over the last months and years, we have seen a flurry of advances in LLM circuit discovery methods. These methods involve both automated (Conmy et al., 2023) and manually discovered circuits (Wang et al., 2022c; Marks et al., 2025). These methods make sweeping statements about the functional roles of particular heads in an LLM based on the

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Preprint. February 20, 2026.

patterns emerging in their attention score matrices (Elhage et al., 2021; Wang et al., 2022b; Olsson et al., 2022; Voita et al., 2019). However, we know that multiple theoretical solutions can exist for the same problem. For example, transformers have been shown to model "backup" circuitry that is utilized only when the primary circuit is compromised. And proxy models for interpretability such as transcoders and sparse autoencoders do not learn canonical concepts, even if they have high reconstruction fidelity and are monosemantic (Olah, 2025).

We face a similar problem in the context of LLM circuits and internal representations in general, where we might have different attention head representations for a retrained model but with different parameter initialization (i.e. a different seed). This hinders the application of LLMs in safety-critical areas like finance, nuclear, or healthcare, where the cost of a single point of failure might be huge (Bengio et al., 2025). We cannot rely on features or circuits extracted from a domain-specialized LLM as explanations of the domain's intrinsic mechanisms, because these structures may be unstable across refits. For example, circuits identified in a model trained on disease-specific data should not be treated as a definitive account of the disease without demonstrated consistency across multiple random initializations of the same model. Without these replications, a seemingly "explainable" feature impacting diagnosis may not have any bearing on the true underlying biological mechanisms, and may in fact be entirely spurious (Stanley et al., 2025).

It is therefore crucial to study the "seed stability" of LLM attention heads, the components most critical in circuits. We ask: if two different teams use the same LLM architecture and the same data, do they end up with the same attention heads? While this may be a straightforward question to articulate, comparing two refits of the same model is not trivial. There exist permutation symmetries of hidden units (Ainsworth et al., 2023); that is, one can swap any two units of a hidden layer in a network and, assuming weights are adjusted accordingly, network functionality will not change. Despite the same or similar output prediction performance, these permutations of parameter spaces can cause various transformations of the internal activation space of complex LLMs, which makes the task of comparing among refits even more challenging (Zhang et al., 2025). Cross-seed comparisons made using methods like CKA, CCA, and SVCCA (Kornblith et al., 2019; Hardoon et al., 2004; Raghu et al., 2017) are not flawless and can be misleading (Kornblith et al., 2019; Davari et al., 2022).

Given this, in the present paper, we compare attention head functions using their attention score matrices, which encode token-token relationships on a common basis and are therefore directly comparable across seeds (Kobayashi et al., 2023). This choice yields a simple, linear, and easily attributable metric for assessing the seed stability of attention heads.

Our main contributions and findings based on the experiments detailed in Methods (§ 3) and Results (§ 4) sections are the following:

- **Middle-layer instability**: Attention heads in middle transformer layers are the least stable across refits and the most representationally distinct.
- **Depth dependence**: Instability increases with the depth of the transformer model; alternative instances of deeper models exhibit stronger mid-depth divergence across refits.
- **Functional implication of instability**: With increasing depth of the layer, unstable heads become increasingly influential within their respective layer.
- **Effect of weight decay**: Using AdamW (decoupled weight decay) substantially improves attention-head seed stability across model refits.
- **Residual-stream robustness**: Across model refits, the residual stream is relatively much more stable than the attention heads.

From our comprehensive experiments, we discover that the seed stability of a GPT-like architecture is clearly tied to the various choices we make while deciding upon the training configuration of the model (e.g., optimiser, weight decay). These choices as such might have no appreciable impact upon the performance of the final trained model but can have a large impact on the universality of representational structures across model instantiations. This in turn has a cascading effect on the practical usefulness of these representational structures in the broader context of explainability.

2. Related Work

Prior studies exploring variation across refits of Transformer models have largely relied on handpicked task subsets or output-level metrics to assess stability. Similar to our work, these works train multiple seeds of transformer-based architectures, but their emphasis is typically on downstream behavior rather than internal alignment. For example, Wal et al. (2025) trains multiple Pythia variants across nine seeds and primarily assess downstream performance stability and the representational stability of specific linguistic features via probing. Müller-Eberstein et al. (2023) uses MultiBERTs (five BERT refits)(Sellam et al., 2022) to conduct longitudinal analyses across training checkpoints, focusing on the evolution of an internal task subspace. Similarly, Michaelov et al. (2025) performs a checkpoints-wise

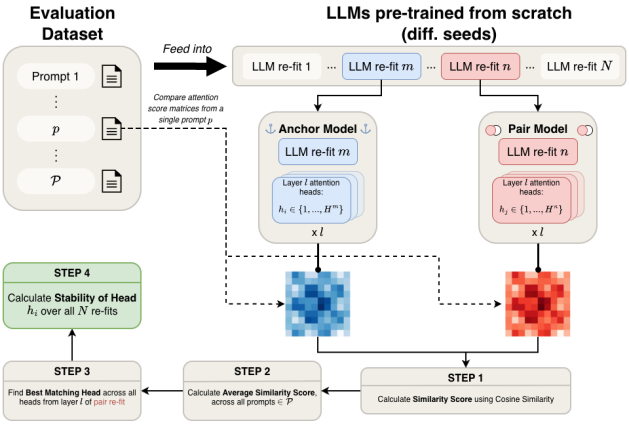

Figure 1. Experimental workflow for assessing attention head stability. For each of our 26 architectural configurations, we instantiate N GPT-2-small LLMs, each with a different initial random seed. Each LLM is pre-trained from scratch with the OpenWebText or C4 datasets, depending on the architectural configuration. After pre-training, we take $\mathcal{P}$ evaluation prompts and compute attention score matrices for each head in each layer. An anchor model m is selected and compared to a pair model n, recursively for each randomly initialized and pre-trained LLM in that architectural subset. Cosine similarity is used to assess the similarity of each attention score matrix from each head, and a "Stability" score is computed from the best matching heads across refits of the architectural subset. See Methods § 3.2 for definitions of these terms.

longitudinal study over six seeds for three Parc families (Parc-Pythia, Parc-Mamba, Parc-RWKV). Notably, most of these efforts are contingent on architectures other than the decoder-only transformer defined in Vaswani et al. (2023). While Karamcheti et al. (2021) reports training five GPT-family seeds at multiple scales, the paper does not analyze internal representations across refits.

By contrast, our methods performs inter-seed comparison in GPT-like (decoder-only) architectures spanning scales from 2-layer (7M parameters) to 12-layer GPT-2-small (124M parameters; (Radford et al., 2019)). Unlike prior LLM-XAI work, we compare internal representations directly and in a task-agnostic manner by operating on attention score matrices. To the best of our knowledge, no previous work has compared attention heads with their representations across seeds in decoder-only transformers.

3. Methods

3.1. Architectures and training details

We study attention head similarity in variants of GPT-like (decoder-only) language models. To isolate the intrinsic seed-stability behavior, we hold most training choices fixed at standard defaults, varying only fundamental architectural factors—depth (2, 4, 8) and heads per layer (8 or 16). Following Elhage et al. (2021), we also trained identical attention-only variants of the above architectures. In addition to this, we also trained a *GPT2-small* architecture with 12 layers and 12 heads configuration.

Our priority was attribution of seed stability in the context of each model's overall architecture. Many hyperparameters such as learning rate schedules, warm up lengths, etc, were kept constant across refits and architectures. This allowed us to isolate and study the seed-stability as a direct outcome of the architecture itself rather than a combination of minor hyperparameter choices.

The only exception to this was the mirror set of identical variants trained with AdamW (Loshchilov & Hutter, 2019) optimizer instead of Adam (Kingma & Ba, 2015), which is motivated by the "norm growth" rationale discussed in Results section 4.4.1. To probe the effect of over- and under-training regimes on seed stability (Hoffmann et al., 2022), we also varied the pretraining corpus and size—the 12 layered GPT2-small models were trained on a bigger 9 billion tokens of OpenWebText (Gokaslan et al., 2019) dataset, whereas the 2, 4, and 8 layered models were trained on a smaller 2 billion token subset of C4 (Nanda, 2022; Raffel et al., 2020) dataset.

In total, we focused on **26 architectures**. We trained 50 refits each for 2, 4, and 8 layered variants and 5 refits each for 12 layered (GPT2-small) architectures. Associated code and complete training/architecture configuration templates are provided in App. A.2, A.3 and the Supplementary Material. Consistent results across these architectural & dataset variations (§ 4) strengthen our analysis & interpretation.

3.2. Attention-head stability across refits

We quantify a concrete notion of stability of an attention head by measuring how consistently it can be matched to a head in the **same** layer of independently trained refits of the same architecture. Let there be N refits (same architecture, hyperparameter set and data, yet different random seeds).

Setup and notation

- Let $\mathcal{P}$ be a fixed set of prompts.
- Choose an *anchor* refit m and a *pair* refit n ($n \neq m$).
- h_i is a head belonging to layer l in *anchor* refit m.
- h_j can be any head belonging to layer l in *pair* refit n.
- All possible h_j heads are considered as *candidate heads* to be compared & matched with h_i head.

3.2.1. Step 1: Prompt-wise head similarity

For a head h_i of layer l in anchor refit m and one of its candidate heads h_j in pair refit n, we define the prompt-wise similarity between h_i and h_j as the *cosine similarity* between their vectorized (flattened) post-softmax attention score matrices for prompt $p \in \mathcal{P}$, that is $(\big(A^m_{h_i}(p)\big), \big(A^n_{h_j}(p)\big))$, from